GLA UNIVERSITY, MATHURA

B.Tech. II Year, III/IV Semester

Subject Name- Probability and Statistics

Practice Sheet – Discrete and Continuous Distribution

Ques 1. If I were explaining to my 6-year-old goddaughter what I am learning in school regarding binomial and Poisson probability distribution, I would try to explain it in a way that makes sense and is relatable. For instance, when explaining binomial probability distribution. If 2% of the items of a factory are defective. The items are packed in boxes of 100 pieces. Using Binomial and Poisson distributions find there will be

i) Two defective items.
ii) At least three defective items.

Ques 2. Students arrive at a local bar and restaurant according to an approximate Poisson process at a mean rate of 30 students per hour. What is the probability that the bouncer has to wait more than three minutes to card the next student?

Ques 3. A grinding machine is so set that its production shafts have an average diameter of 10.10 cm and a standard deviation of 0.20 cm. The product specifications call for shaft diameters between 10.05 cm and 10.20 cm. what proportion of output meets the specifications presuming normal distribution? [$P(-0.25<z<0.5) = 0.2902$]

Ques 4. X is a normally distributed variable with mean $\mu = 30$ and standard deviation $\sigma = 4$. Find the probabilities [given that $P(z<2.5) = 0.9938$, $P(z>-2.25) = 0.9878$, $P(z<1.25) = 0.8944$)

a. $P(X < 40)$
b. $P(X > 21)$
c. $P(30 < X < 35)$

Ques 5. A radar unit is used to measure the speeds of cars on a motorway. The speeds are normally distributed with a mean of 90 km/hr and a standard deviation of 10 km/hr. What is the probability that a car picked at random is traveling at more than 100 km/hr? given that $P(z<1) = 0.8413$

Ques 6. For certain types of computers, the length of time between charges of the battery is normally distributed with a mean of 50 hours and a standard deviation of 15 hours. John owns one of these computers and wants to know the probability that the length of time will be between 50 and 70 hours. Given that $P(z<1.33) = 0.9082$

Ques 7. Entry to a certain University is determined by a national test. The scores on this test are normally distributed with a mean of 500 and a standard deviation of 100. Tom wants to be admitted to this university and he knows that he must score better than at least 70% of the students who took the test. Tom takes the test and scores 585. Will he be admitted to this university? $P(z<0.85) = 0.8023$

Ques 8. The length of similar components produced by a company is approximated by a normal distribution model with a mean of 5 cm and a standard deviation of 0.02 cm. If a component is chosen at random.

a) what is the probability that the length of this component is between 4.98 and 5.02 cm?
b) What is the probability that the length of this component is between 4.96 and 5.04 cm?

Ques 9. At a particular supermarket, 85 percent of customers have the supermarket clubcard.

a) Determine the probability that, in a random sample of 5 customers exactly 4 customers have the supermarket Clubcard.
b) Determine the probability that, in a random sample of 40 customers more than 30 customers have the supermarket Clubcard.

Ques 10. The proportions of people with blood groups O, A, B and AB in a particular population are in the ratio 48:35:12:5, respectively.

a) Determine the probability that a random sample of 20 people from the population contains.
   i) Exactly 10 with blood group O.
   ii) At most 2 with blood group AB.
   iii) At least 8 with blood group A.

Ques 11. Twenty percent of coloured beads used in costume jewellery are blue.

a) Determine the probability that
   i) In a string of 20 beads, more than 3 beads are blue.
   ii) In a string of 28 beads, exactly 4 beads are blue.

Ques 12. The random variable X has a binomial distribution B(11, p). If $P(X = 8) = P(X = 7)$. Find the value of p.

Ques 13. A random variable T has the distribution Geo(1/5). Find

i) P (T = 4)
ii) P (T > 4)
iii) E(T)
iv) V(T)

Ques 14. Andrea practises shots at goal. For each shot the probability of her scoring a goal is 2/5. Each shot is independent of other shots.

i) Find the probability that she scores her first goal.
   a) On her 5th shot,
   b) Before her 5th shot.

Ques 15. In the manufacture of glassware, bubbles can occur in the glass which reduces the status of the glassware to that of a "second". If, on average, one in every 1000 items produced has a bubble, calculate the probability that exactly six items in a batch of three thousand are seconds.